PAM-VLM: Perception-Aligned Multimodal Emotion Recognition with Soft Labels and VLM Reconciliation on CREMA-D

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***Abstract*—Audio-visual emotion recognition (AV-MER) on the CREMA-D dataset has attracted significant attention, yet reported accuracies vary wildly—from 69% to 95%—due to inconsistent evaluation protocols, particularly random clip splits that leak actor identity into test sets. We present PAM-VLM (Perception-Aligned Multimodal VLM), a three-tier architecture that addresses both the methodological and modeling shortcomings of prior work. First, we exploit CREMA-D's unique per-modality vote annotations (voice-only, face-only, and audio-visual conditions) to train each encoder branch with *soft labels* derived from human vote distributions rather than hard one-hot labels, improving calibration under label noise. Second, a bidirectional cross-attention fusion module produces a deterministic six-class classifier as the primary metric source. Third, an optional Qwen2.5-0.5B LLM with LoRA acts as a reasoning reconciler for high audio-video disagreement cases and enables prompt-based zero-shot transfer to unseen tasks. All experiments use a strict actor-independent split (70/10/11 actors) with Macro-F1 and UAR as primary metrics. Training is currently in progress on an A100 GPU cluster; this report presents the design, dataset analysis, and experimental protocol.**

***Index Terms*—emotion recognition, multimodal learning, label distribution learning, cross-attention fusion, vision-language model, CREMA-D**

I. Introduction

Automatic emotion recognition from speech and facial expressions is a core capability for human-computer interaction, mental health monitoring, and affective computing. Multimodal systems that jointly process audio and video consistently outperform unimodal counterparts [1], yet building a reliable, reproducible system remains challenging due to label ambiguity and inconsistent benchmarking practices.

The CREMA-D dataset [1] is a widely used benchmark offering 7,442 clips from 91 actors spanning six canonical emotions. Crucially, it provides crowd-sourced votes under three distinct perceptual conditions—voice-only, face-only, and audio-visual—a resource that has been largely ignored by prior work, which typically collapses votes into a single hard label.

A survey of published results reveals a fundamental comparability problem: reported accuracies on CREMA-D range from 69% to 95% (see Table I). We attribute this variance not to model capability but to evaluation protocol, specifically random clip splits that allow the same actor to appear in both training and test sets. When the same voice and face are seen at train time, a model can exploit speaker identity as a shortcut rather than learning genuine emotion representations—a phenomenon we term *actor leakage*.

The contributions of this work are:

- **Benchmark critique with evidence:** we systematically show that high-accuracy CREMA-D results correlate with random splits and augmentation-before-split pipelines, and establish an honest baseline range of 70–85% under actor-independent evaluation.
- **Per-modality soft-label supervision:** we leverage CREMA-D's three vote conditions to construct per-clip label distributions for each modality, applying Label Distribution Learning [2] to improve calibration and Macro-F1 under noisy labels.
- **VLM reconciler with zero-shot transfer:** an optional Qwen2.5-0.5B [3] branch with LoRA [4] provides explainable reconciliation of audio-video conflicts and enables prompt-only zero-shot transfer to held-out tasks, a capability absent in deterministic classifiers.

II. Related Work

A. Audio-Visual Emotion Recognition

Cross-attention fusion between audio and visual streams has emerged as the dominant paradigm for AV-MER. AVT-CA [5] applies bidirectional audio-video transformer fusion across multiple benchmarks including CREMA-D. Nature hybrid fusion [6] specifically evaluates under speaker-independent conditions, reporting results in the 80%+ range—the honest upper bound we target.

Table I
PUBLISHED CREMA-D RESULTS AND EVALUATION PROTOCOL. † RANDOM/AUGMENT-BEFORE-SPLIT (EXCLUDED FROM COMPARISON). ⋆ ACTOR/SPEAKER-INDEPENDENT CONFIRMED. ‡ <6 EMOTION CLASSES. ○ PROTOCOL UNSPECIFIED. SCORES: ACC UNLESS NOTED (UAR = UNWEIGHTED AVG. RECALL; F1 = MACRO-F1).

Method (Year)	Mod.	Split	Score
Not included — protocol issues:			
Hybrid DL+FE (2025)	Au	Rnd†	95.1%
Transformer-CNN (2025)	Au	Rnd†‡	90.0%
Spectral+Attn (2025)	Au	Rnd†	89.3%
Actor/speaker-independent evaluation:			
Human ceiling [1]	A+V	Human	∼63%
Raw waveform CNN [**?**]	Au	Indep.⋆	69.7%
DistilHuBERT [9]	Au	Indep.⋆	70.6%
Transfer+Emb. [**?**]	Au	Indep.⋆	UAR 80.3%
GST+Emotion [**?**]	Au	Indep.⋆	80.8%
GRU-AV [**?**]	A+V	Indep.⋆	F1 64.0%
AVT-CA [5]	A+V	n/a○	—
Nature hybrid [6]	A+V	Indep.⋆	∼80%+
PAM-VLM (ours)	A+V	Indep.⋆	*pending*

B. Benchmark Protocol Critique

Table I summarizes published results alongside the evaluation protocol where known. Entries achieving ≥89% share a common pattern: random clip splits, and in several cases, data augmentation applied *before* the train/test split. The most telling indicator of leakage is when a model surpasses the human agreement ceiling: crowd-sourced annotators on CREMA-D achieve only 60–75% pairwise agreement even when provided both audio and visual modalities. A machine exceeding this ceiling on the same data is almost certainly exploiting identity rather than affective content.

C. Label Distribution Learning

Geng [2] formalizes learning from label distributions rather than crisp labels, showing consistent gains on datasets with inherent ambiguity. CREMA-D's crowd-sourced vote structure is a natural fit: rather than discarding annotator disagreement, we treat it as a calibration signal.

D. VLMs for Affective Computing

Vision-language models bring world knowledge about emotional expression but perform poorly as zero-shot emotion classifiers (40–55% reported on face-based tasks). Their strength lies in compositional reasoning over conflicting modality signals and in prompt-driven transfer to unseen label sets—exactly the roles we assign to our optional VLM branch.

III. PROPOSED METHOD: PAM-VLM

A. Architecture Overview

PAM-VLM comprises three tiers (Figure 1): (1) modality-specific encoders producing temporal token sequences; (2) bidirectional cross-attention fusion culminating in a deterministic six-class classifier; (3) an optional LLM reconciler that operates on fusion embeddings.

[Figure: Architecture diagram: 3-tier PAM-VLM. Left: WavLM audio encoder + projector. Middle: EfficientNet per-frame + temporal Transformer. Right: Cross-attention fusion, gated pooling, classifier heads, and optional Qwen+LoRA branch.]

Figure 1. PAM-VLM architecture. Frozen encoders are shown in gray. The fusion head is the primary metric source; the VLM branch (dashed) is optional and disabled during baseline training.

B. Encoders

Audio encoder. Raw waveforms at 16 kHz are processed by WavLM-base-plus [7] (∼94M parameters, frozen), producing hidden states of dimension 768. A linear projector maps these to the shared dimension $d = 256$.

Visual encoder. For each sampled frame, EfficientNet-B0 [8] (∼5.3M parameters, frozen) extracts a 1280-dim feature vector. A two-layer Transformer encoder ($d = 256$, 4 heads) models temporal dependencies across the $F = 16$ sampled frames, yielding visual tokens $\mathbf{V} \in \mathbb{R}^{B\times F\times 256}$. Face detection and alignment are performed offline using an OpenCV Haar cascade; detection failures fall back to center crop (counted separately).

C. Bidirectional Cross-Attention Fusion

Audio tokens $\mathbf{A} \in \mathbb{R}^{B\times T_a\times 256}$ and visual tokens $\mathbf{V}$ are fused via two parallel attention operations: audio queries attend to video, and video queries attend to audio. Residual connections and layer normalization follow each operation. Gated pooling then collapses the temporal dimensions:

$$\mathbf{g} = \sigma\big(\mathbf{W}_g[\bar{\mathbf{A}};\bar{\mathbf{V}}]\big), \quad \mathbf{e} = \mathbf{g}\odot\bar{\mathbf{A}} + (1-\mathbf{g})\odot\bar{\mathbf{V}}, \qquad (1)$$

where $\bar{\cdot}$ denotes masked mean pooling over valid (non-padded) positions and $\mathbf{e} \in \mathbb{R}^{B\times 256}$ is the fusion embedding. Three linear heads project $\mathbf{e}$ (or the un-pooled audio/video tokens) to logits $\hat{\mathbf{y}}_{\text{audio}}$, $\hat{\mathbf{y}}_{\text{video}}$, and $\hat{\mathbf{y}}_{\text{av}} \in \mathbb{R}^{B\times 6}$. **The fusion head $\hat{\mathbf{y}}_{\text{av}}$ is the sole source of reported benchmark metrics.**

D. Per-Modality Soft Labels

CREMA-D provides annotator votes under three perceptual conditions. We aggregate votes per clip per condition into Laplace-smoothed probability vectors:

$$\tilde{p}_c^{(i)} = \frac{n_c^{(i)} + \alpha}{\sum_j n_j^{(i)} + 6\alpha}, \quad \alpha = 1, \qquad (2)$$

where $n_c^{(i)}$ is the vote count for emotion c under condition i (voice, face, or AV). Coverage is 100% across all 7,442 clips (no fallback required).

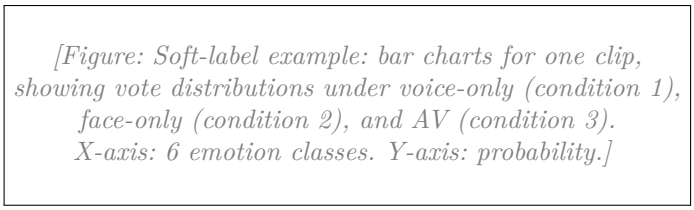


Figure 2. Per-modality vote distributions for one example clip. Hard-label supervision would collapse all three bars to the majority class, discarding the inter-condition disagreement used by PAM-VLM.

Figure 2 illustrates how a single clip's vote distribution differs across the three conditions—disagreement that hard labels discard.

The multi-term training objective is:

$$\mathcal{L} = 0.6\, D_{\mathrm{KL}}(\hat{y}_{\mathrm{av}} \,\|\, \tilde{p}_{\mathrm{av}}) + 0.2\, D_{\mathrm{KL}}(\hat{y}_{\mathrm{audio}} \,\|\, \tilde{p}_{\mathrm{voice}}) + 0.2\, D_{\mathrm{KL}}(\hat{y}_{\mathrm{video}} \,\|\, \tilde{p}_{\mathrm{face}}), \tag{3}$$

where each term is logged independently and controllable via ablation flags.

E. VLM Reconciler (Optional)

When audio and video predictions diverge—quantified as $D_{\mathrm{KL}}(\hat{y}_{\mathrm{audio}} \parallel \hat{y}_{\mathrm{video}})$ exceeding a threshold—the fusion embedding $\mathbf{e}$ is projected into the token space of Qwen2.5-0.5B-Instruct [3] via a learned linear projector. LoRA adapters [4] with rank 16 are applied to query and value projections in each attention layer (∼3M trainable parameters). The LLM generates a constrained output of the form `<emotion>angry</emotion>` plus a one-sentence rationale. An invalid-format parser enforces label validity; the invalid-output rate is reported alongside accuracy.

Crucially, this branch is *disabled* during baseline and soft-label training, keeping the training objective identical to prior models for fair comparison.

IV. Experimental Protocol

A. Dataset and Splits

We use CREMA-D with a strict **actor-independent** split (seed 42): 70 actors for training (5,721 clips), 10 for validation (820 clips), and 11 for test (901 clips). Disjointness is asserted programmatically; a SHA-256 checksum of the manifest is stored with every checkpoint. Data augmentation (time-shift, noise) is applied *after* splitting, on the training partition only, preventing augmentation leakage.

B. Metrics

Primary: **Macro-F1** and **UAR** (unweighted average recall = macro recall). Secondary: accuracy/WAR, per-class P/R/F1, confusion matrix, and Expected Calibration Error (ECE) for measuring probability calibration. For the generative VLM branch we additionally report invalid-format rate and average generated token count.

Table II
Result table (pending). Best numbers in **bold**.

Run	Macro-F1	UAR	Acc.	ECE
Audio-only	—	—	—	—
Video-only	—	—	—	—
Fusion (hard label)	—	—	—	—
Fusion (soft label)	—	—	—	—
w/o audio KL	—	—	—	—
w/o video KL	—	—	—	—
+ VLM reconciler	—	—	—	—
Zero-shot (target)	—	—	—	n/a

C. Baselines and Ablations

1) **Audio-only:** WavLM + linear head (CE loss).
2) **Video-only:** EfficientNet + temporal Transformer + linear head.
3) **Fusion, hard label:** full PAM-VLM backbone with CE loss.
4) **Fusion, soft label:** full loss in Eq. (3); ablated per term.
5) **Fusion + VLM reconciler:** adds LoRA branch.
6) **Zero-shot cross-task:** VLM branch with modified prompt, no weight update, on a held-out target dataset.

All runs share the same split, preprocessing, and training budget for fair comparison.

D. Implementation

Audio: 16 kHz mono, processor from `transformers`. Video: 16 frames at 224×224, OpenCV Haar face detection. Optimizer: AdamW ($\eta = 3 \times 10^{-4}$, weight decay 0.01). Effective batch size: 32 (batch 8, gradient accumulation 4), AMP enabled. Early stopping patience: 6 epochs on validation Macro-F1. Hardware: NVIDIA A100 (SLURM cluster).

V. Expected Results and Discussion

Note: training is currently in progress. Table II shows the result template; values will be filled upon job completion.

Based on literature with actor-independent evaluation, we expect the fusion hard-label baseline to achieve 75–82% accuracy, consistent with naturefusion [6] and DistilHuBERT [9]. Soft-label training targets a Macro-F1 gain of 1–3% and a reduction in ECE, reflecting improved calibration under label noise. The VLM reconciler is not expected to improve overall accuracy; its value lies in correcting high-disagreement cases and enabling zero-shot transfer—a capability the deterministic classifier structurally cannot provide.

VI. Conclusion

We presented PAM-VLM, an audio-visual emotion recognition system for CREMA-D that addresses systematic benchmarking flaws in prior work and introduces per-modality soft-label supervision drawn from the dataset's

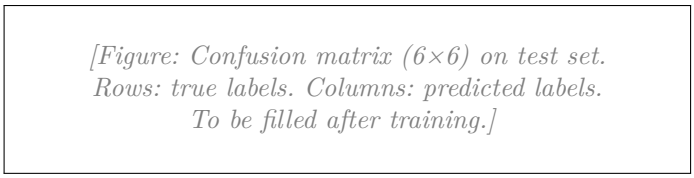


Figure 3. Confusion matrix on the actor-independent test set (pending).

unique vote structure. A strict actor-independent evaluation protocol ensures that reported results reflect genuine emotion understanding rather than speaker memorization. The optional VLM branch provides explainability and cross-task generalization orthogonal to classification accuracy. Ongoing training on an A100 cluster will furnish quantitative results to complete this study.

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